Lecture 12

Maxima and Minima

Example 1: $f(x) \quad = x^3 + 3x^2 - 9x + 1$

i. Find the open intervals on which f is increasing.

ii. Find the open intervals on which f is decreasing.

iii. Find the open intervals where f is concave up.

iv. Find the open intervals where f is concave down.

v. Find the x-coordinate of all inflection points.

vi. Locate the critical numbers and identify which numbers correspond to stationary points.

Solution:

$(a)\ f(x) = x^3 + 3x^2 - 9x + 1,$

$$\begin{aligned} \text{So, } f'(x) &= 3x^2 + 6x - 9 \\ &= 3(x^2 + 2x - 3) \\ &= 3\{x(x+3) - 1(x+3)\} \\ &= 3(x+3)(x-1) \end{aligned}$$

For the critical points: (Stationary Points)

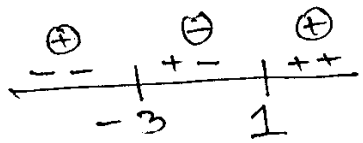
$f'(x) = 0$

$\Rightarrow 3(x+3)(x-1) = 0$

$\Rightarrow (x+3)(x-1) = 0$

Either $(x+3) = 0$ Or, $(x-1) = 0$

$\Rightarrow x = -3$ $\Rightarrow\ x = 1$



Now,

Interval	$\mathbf{3(x+3)(x-1)}$	$\boldsymbol{f'(x)}$	**Conclusion**
$x < -3$	$(-)(-)$	$(+)$	$f'(x)$ is positive. So, $f(x)$ is increasing on $(-\infty, -3)$.
$-3 < x < 1$	$(+)(-)$	$(-)$	$f(x)$ is decreasing on $(-3,1)$.
$x > 1$	$(+)(+)$	$(+)$	$f(x)$ is increasing on $(1, +\infty)$.

i. $f(x)$ is increasing on $(-\infty, -3)$ and $(1, +\infty)$.

ii. $f(x)$ is decreasing on $(-3,1)$.

vi. The critical points and stationary points are the same for $f(x)$. Hence, the critical and stationary points are, $x = -3,1$.

<u>Second Derivative</u>:

$f''(x) = 6x + 6 = 6(x + 1)$

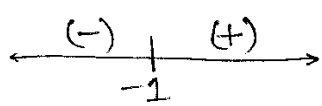
For inflection point,

$f''(x) = 0$

$\Rightarrow 6x + 6 = 0$

$\Rightarrow 6x = -6$

So, $x = -1$.



Now,

Interval	$\mathbf{6(x+1)}$	$\boldsymbol{f''(x)}$	**Conclusion**
$x < -1$	$(-)$	$(-)$	$f(x)$ is concave down on $(-\infty, -1)$.
$x > -1$	$(+)$	$(+)$	$f(x)$ is concave up on $(-1, +\infty)$.

Therefore:

iii. $f(x)$ is concave up on $(-1, +\infty)$.

iv. $f(x)$ is concave down on $(-\infty, -1)$.

v. The inflection point is at $x = -1$.

H.W. Do (i-vi) for the following functions

1. $f(x) = 12x^3 + 12x^2 - 24x$
2. $f(x) = x^3 + x^2 - 2x$
3. $f(x) \quad = 5 + 12x - x^3$

Example 2: $f(x) = x^{\frac{2}{3}}$. Locate the critical numbers and identify which critical numbers correspond to stationary points.

Solution:

$f(x) = x^{\frac{2}{3}}$

$\Rightarrow f'(x) = \frac{2}{3}x^{\frac{2}{3}-1}$

$\Rightarrow f'(x) = \frac{2}{3}x^{-\frac{1}{3}} = \frac{2}{3}\left(\frac{1}{x^{\frac{1}{3}}}\right) = \frac{2}{3(x^{1/3})}$

Here, if $x^{1/3} = 0 \Rightarrow x = 0$.

So, for $x = 0$, $f'(x) = \frac{2}{3(x^{1/3})}$ is undefined.

$\therefore x = 0$ is a critical point.

Again, $f'(x) = 0$

$\Rightarrow \frac{2}{3(x^{1/3})} = 0$

$\Rightarrow \frac{2}{3} = 0$

It is not possible since $\frac{2}{3} \neq 0$.

So, we cannot have any point for which $f'(x) = 0$. Therefore, we cannot have any stationary point.

H.W. Practice sheet (1,2)

Relative Maximum and Minimum (Relative Extrema):

(i) Relative Maximum:

A function f is said to have a relative maximum at $x = x_0$, if there is an open interval containing x_0 on which $f(x_0)$ is the largest value, that is, $f(x_0) \geq f(x)$ for all x in the interval.

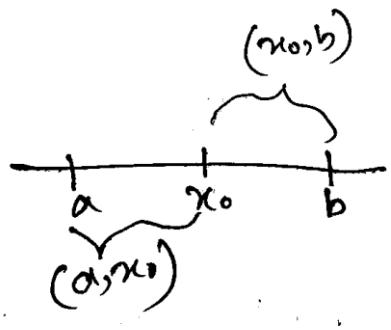
(ii) Relative Minimum:

A function f is said to have a relative minimum at $x = x_0$, if there is an open interval containing x_0 on which $f(x_0)$ is the smallest value. That is, $f(x_0) \leq f(x)$ for all x in the interval.

(iii) Relative Extremum:

If f has either a relative maximum or a relative minimum at $x = x_0$, then f is said to have a relative extremum at x_0.

First Derivative Test:



(i) If $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a relative maximum at x_0.

(ii) If $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a relative minimum at x_0.

(iii) If $f'(x)$ has the same sign on both (a, x_0) and (x_0, b), then f does not have a relative extremum at x_0.

Second Derivative Test:

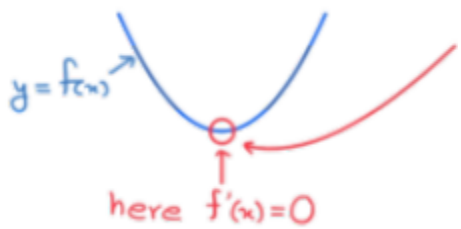
Let, f is twice differentiable at a point $x = x_0$,

(i) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at $x = x_0$.

(ii) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at $x = x_0$.

(iii) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive. That is, f may have a relative maximum, a relative minimum, or neither at x_0. It might even have a point of inflection.

❖ Recall that $f''(x) > 0$ in a region means the curve is concave up in that region.



We can see that the turning point in a curve that is concave up is a minimum point.

❖ Similarly, $f''(x) < 0$ in a region means the curve is concave down in that region.

We can see that the turning point in a curve that is concave down is a maximum point.

These turning points are said to be the relative maximum or relative minimum points, or simply relative extrema.

When we cannot determine whether the turning point is a maximum or a minimum from the second derivative test, we use the first derivative test.

Example: Let $f(x) = 2x^3 - 9x^2 + 12x$. Find the relative extrema (maxima and minima) using both the first and second derivative tests. Find the maximum and minimum values.

Solution:

<u>First Derivative Test</u>

Given, $f(x) = 2x^3 - 9x^2 + 12x$.

$$\begin{aligned} f'(x) &= 6x^2 - 18x + 12 \\ &= 6(x^2 - 3x + 2) \\ &= 6(x^2 - 2x - x + 2) \\ &= 6\{(x-2)(x-1)\} \end{aligned}$$

For stationary points (critical points):

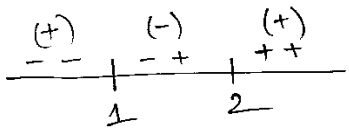
$f'(x) = 0$

$\Rightarrow 6(x-2)(x-1) \quad = 0$

$\Rightarrow (x-2)(x-1) \quad = 0$

Either $(x-2) = 0$ or $(x-1) = 0$

$\Rightarrow x = 2 \qquad$ or $\qquad x = 1$



Interval	$\mathbf{6(x-2)(x-1)}$	$\boldsymbol{f'(x)}$
$x < 1$	$(-)(-)$	$(+)$
$1 < x < 2$	$(-)(+)$	$(-)$
$x > 2$	$(+)(+)$	$(+)$

So, $f(x)$ has relative maximum at $x = 1$.

$f(x)$ has relative minimum at $x = 2$.

Second Derivative Test

Now, $f'(x) = 6x^2 - 18x + 12$.
Then, $f''(x) \quad = 12x - 18$.

At $x = 1$:

$f''(1) \quad = 12(1) - 18 = -6 < 0$

$\therefore f(x)$ has a relative maximum at $x = 1$.

At $x = 2$:

$f''(2) \quad = 12(2) - 18 = 24 - 18 = 6 > 0$

$\therefore f(x)$ has a relative minimum at $x = 2$.

Minimum value, $f(2) \quad = 2(2)^3 - 9(2)^2 + 12(2) = 16 - 36 + 24 = 4$

Maximum value, $f(1) \quad = 2(1)^3 - 9(1)^2 + 12(1) = 2 - 9 + 12 = 5$

H.W. Find the maximum and minimum values of the function: $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$.

Answer: Maximum at $x = 2$, and the value is -3.
Minimum at $x = 1,3$, and the values are -4 and -4.

Example: Use the given derivative to find all critical numbers of f and determine whether a relative maximum, relative minimum, or neither occurs there: $f'(x) = \frac{x^2-1}{x^2+1}$.

Solution: Given, the first derivative,

$$f'(x) = \frac{x^2 - 1}{x^2 + 1}$$

Here, for all real values of x, $x^2 + 1 > 0$.

So, there is no value of x for which $f'(x)$ is undefined.

Now, $f'(x) = 0$

$$\Rightarrow \frac{x^2 - 1}{x^2 + 1} = 0$$

$$\Rightarrow x^2 - 1 = 0$$

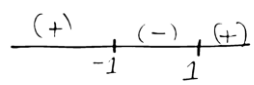
$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

So, $x = 1, x = -1$

Here, the critical points and stationary points are same. So, $x = 1, x = -1$ are the critical points and stationary points.

First Derivative Test



Interval	$\frac{x^2-1}{x^2+1}$	$f'(x)$
$x < -1$	$(+)$	$(+)$
$-1 < x < 1$	$(-)$	$(-)$
$x > 1$	$(+)$	$(+)$

So, $f(x)$ has relative maximum at $x = -1$.

$f(x)$ has relative minimum at $x = 1$.

H.W. Let, $f(x) = 17 - 15x + 19x^2 - x^3$. Find the critical points, stationary points, intervals for increasing, decreasing, concave up and concave down regions and also find maximum and minimum value.

Example: Find the relative extrema of $f(x) = 3x^5 - 5x^3$ using the first derivative test.

Solution: Given, $f(x) = 5x^5 - 5x^3$.

Now, $f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$

For critical points, $f'(x) = 0$

$\Rightarrow 15x^2(x^2 - 1) = 0$

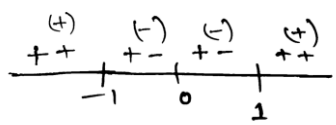
Either $15x^2 = 0$ or $(x^2 - 1) = 0$

$\Rightarrow x^2 = 0$ or $x^2 = 1$

$\Rightarrow x = 0$ or $x = \pm 1$

So, critical points at $x = 0$, $x = 1$ and $x = -1$.

In the first derivative test, we check where the function is increasing or decreasing.



Interval	$\mathbf{15x^2(x^2 - 1)}$	$\boldsymbol{f'(x)}$
$x < -1$	$(+)(+)$	$(+)$
$-1 < x < 0$	$(+)(-)$	$(-)$
$0 < x < 1$	$(+)(-)$	$(-)$
$x > 1$	$(+)(+)$	$(+)$

For the turning point $x = -1$, to its left $(x < -1)$, $f'(x)$ is increasing. To its right $(-1 < x < 0)$, $f'(x)$ is decreasing.

So, $f(x)$ has relative maximum at $x = -1$.

For the turning point $x = 1$, to its left $(0 < x < 1)$, $f'(x)$ is decreasing. To its right $(x > 1)$, $f'(x)$ is increasing.

So, $f(x)$ has relative minimum at $x = 1$.

At $x = 0$, to its left $(-1 < x < 0)$, $f'(x)$ is negative, means it is decreasing.

To its right $(0 < x < 1)$, $f'(x)$ is negative, means it is decreasing.

So, it's not a maximum nor a minimum. Therefore, it's a point of inflection.

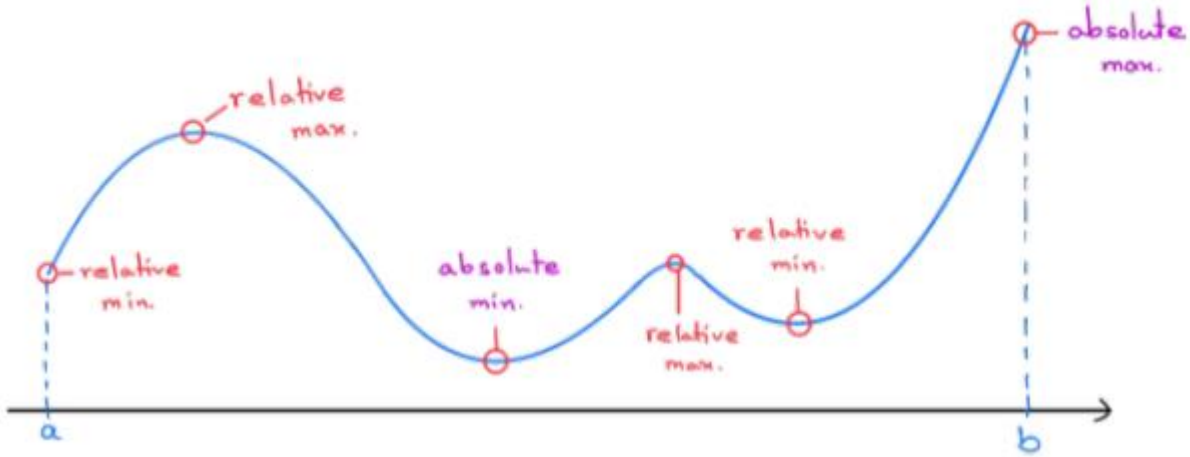
At $x = 0$, we have $f''(x)$=0. So, we can't determine its nature using the second derivative test.

Absolute Extrema:

Let $f(x)$ be a function defined on an interval $[a, b]$ and let $x_0 \in [a, b]$.

(i) $f(x)$ has an absolute maximum at x_0 if $f(x_0) \geq f(x)$ for all x in the interval $[a, b]$.

(ii) $f(x)$ has an absolute minimum at x_0 if $f(x_0) \leq f(x)$ for all x in the interval $[a, b]$.



H.W. Practice sheet (3,4)